

Attention-Enhanced Vision Transformer for Retinal Disease Classification Using OCT Images

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ABSTRACT

Retinal diseases such as Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen are major causes of vision impairment worldwide. Early diagnosis and treatment are essential to prevent permanent vision loss. Optical Coherence Tomography (OCT) is a non-invasive imaging technique widely used for retinal disease diagnosis. However, manual analysis of OCT images requires expert ophthalmologists and can be time-consuming. This project proposes an automated retinal disease classification system using deep learning techniques. Initially, baseline Convolutional Neural Network (CNN) models such as ResNet50 and EfficientNet-B0 are implemented for performance comparison. To improve classification accuracy, an Attention-Enhanced Vision Transformer (AE-ViT) framework is proposed. The model combines advanced image preprocessing, attention mechanisms, and transformer-based feature extraction to identify retinal abnormalities effectively. The system is evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics. The proposed framework aims to provide a reliable computer-aided diagnostic tool that assists medical professionals in the early detection of retinal diseases and improves healthcare outcomes.

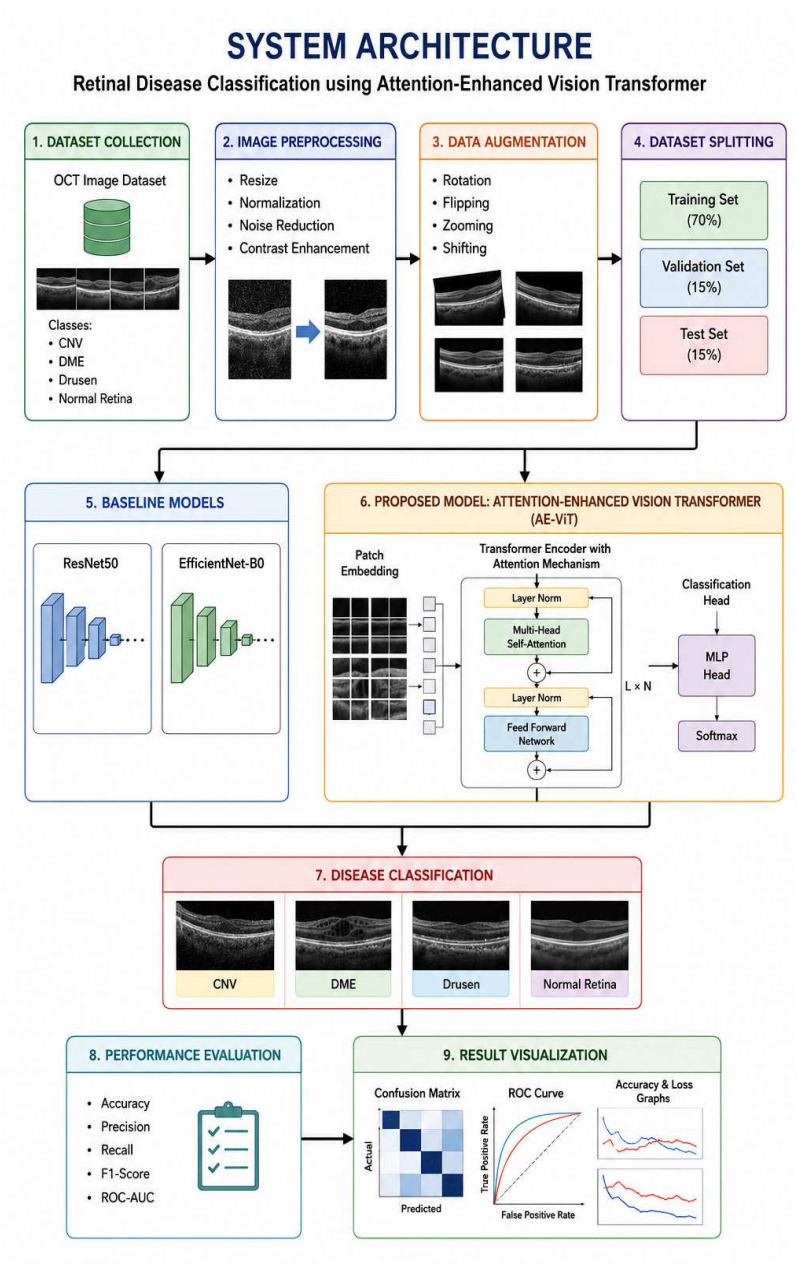
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INTRODUCTION

Artificial Intelligence has significantly transformed the healthcare sector by enabling automated analysis of medical images. Medical image classification helps clinicians diagnose diseases accurately and efficiently, reducing human effort and diagnostic errors. Among various medical imaging modalities, Optical Coherence Tomography (OCT) has become an important tool for diagnosing retinal disorders because it provides high-resolution cross-sectional images of retinal layers. Retinal diseases such as Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen can lead to severe vision impairment if not detected at an early stage. Traditional diagnosis depends heavily on ophthalmologists' expertise and manual image interpretation, which can be time-consuming and subjective. Recent advances in deep learning, especially Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), have demonstrated remarkable success in medical image analysis. Vision Transformers can capture global image features more effectively than conventional CNNs. Furthermore, attention mechanisms help the model focus on disease-specific regions within retinal images.

This project aims to develop an automated retinal disease classification system using an Attention-Enhanced Vision Transformer framework. The proposed system seeks to improve diagnostic accuracy and provide an efficient decision-support tool for ophthalmologists.

SYSTEM ARCHITECTURE



MODULES DESCRIPTION

The proposed Retinal Disease Classification System consists of several modules that work together to identify retinal diseases from OCT images. The first module is **Dataset Collection**, where retinal OCT images are collected from publicly available datasets and categorized into classes such as CNV, DME, Drusen, and Normal Retina. The collected images are verified for quality and proper labeling before being used for training and testing. The second module is **Image Preprocessing**, which prepares the images for analysis by resizing them to a fixed dimension, normalizing pixel values, reducing noise, and enhancing image contrast. This step ensures consistency across the dataset and improves feature extraction. The third module is **Data Augmentation**, which increases the number of training samples through transformations such as rotation, flipping, zooming, and shifting. This helps reduce overfitting and improves the model's ability to generalize to unseen data. The fourth module is **Baseline Model Training**, where deep learning models such as ResNet50 and EfficientNet-B0 are trained on the processed dataset. These models provide baseline results for comparison with the proposed approach. The fifth module is the **Attention-Enhanced Vision Transformer (AE-ViT)**, which serves as the core component of the system. It uses transformer-based architecture along with attention mechanisms to focus on disease-specific retinal regions and extract meaningful features for accurate classification. The sixth module is **Disease Classification**, where the trained model predicts whether an OCT image belongs to CNV, DME, Drusen, or Normal Retina. The generated predictions assist in the early diagnosis of retinal diseases. The seventh module is **Performance Evaluation**, which measures the effectiveness of the models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics. This module helps compare the performance of baseline and proposed models. Finally, the **Result Visualization Module** displays confusion matrices, ROC curves, accuracy graphs, loss curves, and classification results. These visual representations help analyze model performance and validate the effectiveness of the proposed retinal disease classification system.

HARDWARE AND SOFTWARE REQUIREMENTS

Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5 or above
RAM	8 GB Minimum (16 GB Recommended)
Storage	20 GB Free Disk Space
GPU	NVIDIA GTX 1650 or above (Optional)
Display	1366 × 768 Resolution or higher
Internet	Required for dataset download and updates

Software Requirements

Software	Specification
Operating System	Windows 10/11, Ubuntu Linux, macOS
Programming Language	Python 3.10+
IDE	VS Code / PyCharm / Jupyter Notebook
Cloud Platform	Google Colab (Recommended)
Deep Learning Framework	TensorFlow 2.x / PyTorch
Database	Not Required
Version Control	Git & GitHub
Libraries	NumPy, Pandas, OpenCV, Matplotlib, Scikit-Learn, TensorFlow/Keras, PyTorch, Albumentations

CONCLUSION

This project presents an automated retinal disease classification system using Optical Coherence Tomography (OCT) images and deep learning techniques. The proposed system utilizes baseline models such as ResNet50 and EfficientNet-B0 along with an Attention-Enhanced Vision Transformer framework to improve disease classification performance. Through image preprocessing, data augmentation, attention mechanisms, and transformer-based feature extraction, the system aims to accurately classify retinal diseases including CNV, DME, Drusen, and Normal Retina. The performance of the developed model is evaluated using standard metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The proposed approach is expected to achieve higher classification accuracy and better feature representation compared to conventional methods. This system can serve as an effective computer-aided diagnostic tool for ophthalmologists, supporting early disease detection and contributing to improved patient care and clinical decision-making.